COURSE 2 · SECTION 13 · EXERCISE

Test types, ppm by product and the cost of a claim

Section 13 · Product quality & reliability dashboard (portfolio project 2)

Start: Engxtech_S13.pbix (your finished S13 file); save a copy as Engxtech_S13_Exercise.pbix

Time: about 25 min

Data: the course data folder

DESIGN QUESTION

Which validation test type fails most often, is the seat-ring-wear rate the same in every product, and what does a typical claim cost?

Extra practice. This exercise is extra practice: a longer task than the video's short "Your turn" questions, which are answered in the video itself. Do it after the video, in a copy of your file.

In video S13 you built the Quality & reliability page from a blank file: TestResults, Builds, WarrantyClaims and DFMEA with the helper tables Dates, BuildMonths and SeatRingRevs, the matured-cohort columns, and measures such as Validation Pass Rate, Field ppm, CI Low ppm, CI High ppm, Warranty Cost and Seat Ring Warranty Cost. Products links to both Builds and WarrantyClaims, so the ppm measures also work by product.

New in this exercise (not shown in the video; one line each):

- **Per seat ring**: BOM QtyPer for P300: 2 in each pump, 1 in the valve; multiply units by it to get seat rings fitted
- **LOOKUPVALUE**: returns one value from another table, here the P300 QtyPer of a product

Your tasks

1. In your exercise copy, add a table: TestResults[TestType], Lots Tested, Validation Pass Rate. Which test type is lowest?
2. Add a table: Products[ProductID], Wear Failures 90d, Matured Units, Field ppm, CI Low ppm, CI High ppm.
3. Write **Seat Ring ppm**: failures per million seat rings fitted (units x P300 QtyPer of the product). Compare the products again.
4. Write **Avg Seat Claim Cost** and read the warranty cost of all other failure modes.
5. Add a measure for DFMEA lines with RPN After of 100 or more, and name them.
6. Explain why the ppm counts 93 seat-ring-wear claims when there are 207.

Check your answer

Check	Expected
Pass rate, Dimensional	**91.8% (45 of 49)**
Pass rate, Endurance	**89.8% (44 of 49)**
Pass rate, Environmental	**90.9% (60 of 66)**
CP-100: failures / units / ppm (95 % CI)	**50 / 15,726 / 3,179 (2,360 to 4,192)**
BP-200: failures / units / ppm (95 % CI)	**36 / 11,291 / 3,188 (2,233 to 4,414)**
MV-300: failures / units / ppm (95 % CI)	**7 / 6,877 / 1,018 (409 to 2,097)**
ppm per seat ring: CP-100 / BP-200 / MV-300	**1,590 / 1,594 / 1,018**
Average seat-ring-wear claim cost (USD)	**426.31**
Warranty cost, other failure modes (USD)	**6,530**
DFMEA lines with RPN After >= 100	**2 (DF-04, DF-07)**
Seat-ring-wear claims: in ppm / after 90 d / unmatured month	**93 / 110 / 4**

Hint. If every product shows the same ppm, the Products relationship to Builds or to WarrantyClaims is missing. If the cost per claim looks too low, you divided by all 234 claims instead of the 207 seat-ring-wear claims.

Stuck? The worked solution with the full DAX and M is in Solutions/S13_Solution.pdf.